Consider the following linear optimization problem:

$$\min_{x \in \mathbb{R}} 2x_1 + x_2$$

subject to

$$x_1 + x_2 = 3,$$
$$x_1 \geq 0,$$
$$x_2 \geq 0.$$

1. Does the solution $x_1^* = 0$, $x_2^* = 3$ verify all the constraints of the problem? Which constraints are active at (x_1^*, x_2^*)?
2. Answer the same questions with $x_1^* = 2$, $x_2^* = 2$.